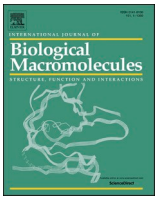
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Effective pretreatment of tea stem via poly-deep eutectic solvent for promoting platform molecule production and obtaining fluorescent lignin

Pei-Ru Zhou [a], Xiao-Ping Zheng [a], Ya-Peng Du [a], Yu Chai [a], Yu-Cang Zhang [a,b,c], Yan-Zhen Zheng [a,b,c,*]

[a] College of Ocean Food and Biological Engineering, Jimei University, Xiamen 361021, China
[b] Fujian Provincial Key Laboratory of Food Microbiology and Enzyme Engineering, Xiamen 361021, China
[c] Research center of food biotechnology of Xiamen city, Xiamen, Fujian 361021, China

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ABSTRACT

In this study, polyethylene glycol 200 (PEG200) was employed as hydrogen bond acceptor, while organic acids served as hydrogen bond donors, to formulate poly-deep eutectic solvents (PDESs), which were utilized to pretreat tea stem. Specially, combining PEG200 and oxalic acid (OA) exhibited a notably high cellulose retention (82.03 %) and most efficient hemicellulose (97.02 %) and lignin removal (70.89 %). The removal of amorphous lignin enhanced the crystallinity of the residues and improved the conversion efficiency of cellulose into levulinic acid. Additionally, the structural alterations in lignin samples were analyzed in comparison to milled wood lignin (MWL). The PEG200-OA system facilitated the cleavage of β-O-4 and β-5 linkages and resulted in the degradation of S-type lignin, accompanied by increased condensation of G units. The resulting lignin displayed a reduced molecular weight (Mw of 1283 g/mol, Mn value 531 g/mol) and nanoscale particle size (D_{50} 212 nm). Furthermore, fluorescent lignin was synthesized through simple oxidation and was used to detect metal ions. Density functional theory (DFT) calculations supported that both PEG200 and OA played a significant role in lignin dissolution, with the weak interactions between DES and lignin primarily driven by hydrogen bonding (characterized as weak, closed-shell, and electrostatic) and van der Waals forces.

1. Introduction

As global fossil fuel reserves dwindle and environmental challenges intensify, the role of bio-refinery technology in efficiently converting lignocellulosic biomass into energy and chemicals becomes increasingly important. Lignocellulose primarily consists of hemicellulose (23 % to 32 %), cellulose (38 % to 50 %), and lignin (15 % to 25 %) [1]. Hemicellulose and cellulose can be hydrolyzed or pyrolyzed to produce various biofuels and biochemicals, including monosaccharides like mannose, glucose, and arabinose [2,3]. Lignin, a complex and amorphous polymer, is derived from phenylpropane units and features a variety of hydrophilic functional groups such as carboxylic acids and hydroxyl groups [4]. This unique structure allows lignin to be used in the production of lignin-based nanoparticles, flocculants, and luminescent materials, among other applications [5–7]. However, the intricate network of covalent and hydrogen bonds linking cellulose, hemicellulose, and lignin poses significant challenges for biomass utilization [8]. Pretreatment of lignocellulose is a crucial step in improving its usability and sustainability [9]. This process alters the biomass's structural components, making it more amenable to conversion into valuable products while reducing waste and meeting the growing demand for high-value commodities [9].

Various pretreatment methods, such as alkaline treatments and organic solvent applications, have been developed to enhance this transformation [10]. However, traditional pretreatment methods often face significant challenges, including equipment corrosion, volatility issues, and the requirement for extreme conditions, all of which pose operational risks and can diminish profitability [11]. Additionally, during many pretreatment processes, lignin is prone to depolymerization and condensation, complicating its recovery and subsequent use. Therefore, the development of more efficient and environmentally friendly pretreatment strategies is crucial for optimizing the economic and ecological benefits of lignocellulose utilization [11].

Deep eutectic solvents (DESs) are gaining great attention as promising, eco-friendly alternatives to traditional solvents for the fractionation of biomass components [12,13]. Notable for their simple synthesis,

* Corresponding author at: College of Ocean Food and Biological Engineering, Jimei University, Xiamen 361021, China.
E-mail address: yanzhenzheng@jmu.edu.cn (Y.-Z. Zheng).

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non-toxicity, and low volatility, DESs are typically formed by combining two or more components that create a eutectic mixture with a melting point lower than that of the individual components [14]. The formation of a network of hydrogen bonds between a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD) enhances the dissolution and separation of biomass components [14]. In the pretreatment of lignocellulosic biomass, DESs have shown a remarkable ability to disrupt the lignin-carbohydrate complex (LCC) and improve lignin dissolution [13,15,16]. This effectiveness has led to the exploration of various DES formulations to overcome biomass recalcitrance in bioenergy production. Additionally, reactive DESs offer the advantage of introducing functional groups such as carbonyl, carboxylic, and ammonium groups during pretreatment [15]. This dual functionality can reduce the need for additional chemicals and energy inputs, presenting a more sustainable approach to biomass conversion.

China is the world's largest producer and consumer of tea. The deep processing of tea into products like tea beverages, instant tea, and tea polyphenols generates substantial amounts of tea stem [17,18]. Disposing of this residue as waste not only squanders valuable resources but also poses environmental risks, contributing to pollution and ecological damage. Tea stem is primarily composed of cellulose, hemicellulose and lignin. Currently, the focus of tea stem utilization in China is largely on extracting active compounds such as tea polyphenols and tea polysaccharides [19]. However, the potential value-added applications of lignin, hemicellulose, and cellulose from tea stem remain largely underexplored. Therefore, pretreating tea stem to effectively separate cellulose, hemicellulose, and lignin is crucial for the sustainable utilization of tea stem resources, offering significant environmental and economic benefits. DESs have been used for the pretreatment of multiple lignocelluloses [20,21], however, there remains a significant opportunity to derive greater value from the extracted fractions of tea stem. Besides, beyond the conversion of carbohydrates into biofuels and biochemicals, further utilization of lignin could enhance both the economic viability and environmental sustainability of lignocellulose-based biorefineries [20,21].

In our study, we present a novel approach for converting tea stem into high-value products. We utilized PDESs composed of PEG200 and OA as an innovative reactive medium for breaking down tea stem, enabling the selective separation of lignocellulosic components. PEG200 serves as a hydrogen bond acceptor and is known for its low viscosity and ability to dissolve a wide range of biomass components, making it an ideal candidate for biomass pretreatment. Organic acids, such as oxalic acid, are efficient hydrogen bond donors, which facilitate the formation of stable DES with PEG200. This specific combination has been shown to enhance the solubility and disrupt the structural integrity of lignocellulosic materials, promoting better pretreatment efficiency. Through various analytical techniques, we assessed the impact of PDES treatment on the biomass structure, confirming its effectiveness in selective delignification and hemicellulose removal. We further enhanced the value of the cellulosic fraction by converting it into levulinic acid (LA), one of the top ten pivotal platform chemicals identified by the U.S. Department of Energy, thus maximizing the utility of cellulose derived from tea stem [22]. This conversion process not only increased the economic potential of the biomass but also confirmed the pretreatment method enhanced cellulose digestibility. The recovered lignin was thoroughly characterized, including analyses of molecular weight, particle size, and structural properties, with comparisons drawn to milled wood lignin (MWL) to underscore the unique attributes resulting from our process. Moreover, the application of the recovered lignin was explored by synthesizing fluorescent lignin via a simple method, which was successfully utilized in detecting Fe^{3+} ions, opening new possibilities for lignin use. Furthermore, density functional theory (DFT) calculations were employed, with the combination of the experimental results to provide a comprehensive understanding of the mechanisms driving efficient lignin extraction. Overall, this study introduced the PDES system composed of PEG200 and oxalic acid for efficiently breaking down tea stem, enabling selective lignocellulose separation and high-value conversion of cellulose into levulinic acid. Additionally, the recovered lignin was valorized as a fluorescent material for Fe^{3+} detection, and DFT analysis provided new insights into the extraction mechanisms, showcasing a sustainable and integrated approach for agricultural byproduct utilization.

2. Materials and methods

2.1. Materials

Tea stem, obtained from a local vendor in Xiamen, China, were first rinsed with deionized water and then subjected to thermal drying at 80 °C to eliminate moisture. After drying, the material was mechanically ground and passed through an 80-mesh sieve to produce a consistent, fine tea stem powder. This powder was directly used in subsequent experiments without further modification. PEG200 was purchased from Aladdin Reagent Co., Ltd., Shanghai, China. Various organic acids, including levulinic acid, oxalic acid, formic acid, and lactic acid, were sourced from Rhawn Reagent Co., Ltd., Shanghai, China. Additionally, the catalyst 1-sulfobutyl-3-methylimidazole p-toluenesulfonate ($[C_4SO_3Hmim][PhSO_3]$), with a purity of over 99 %, was supplied by Shanghai Cheng Jie Chemical Co., Ltd. Other analytical-grade chemicals were procured from Aladdin Reagent Co., Ltd.

2.2. Preparation of deep eutectic solvent

The selected organic acids and PEG200 were mixed in 1:2 M ratio and blended under vigorous stirring at 80 °C, resulting in the formation of translucent DES. After cooling to room temperature, the solution remained clear and uniform, confirming the successful synthesis of the DESs. Using a similar approach, a series of PEG200-OA DESs, varying the molar ratios of HBA/HBD, was prepared.

2.3. Kamlet-Taft solvatochromic parameters measurement

4-Nitroaniline (NH_2) and Nile red (NR) were employed as solvatochromic probes to evaluate the Kamlet-Taft (K-T) polarity and hydrogen bonding capacity of the synthesized PDESs. 50 μL of the dye solution (1.0×10^{-3} M in methanol) was prepared, followed by the removal of methanol through vacuum evaporation at 40 °C for 30 min. The remaining dye residues were then uniformly mixed with 2 g of the respective DES. The absorption spectra of the DES-dye mixtures were recorded using a UV–Vis spectrophotometer, covering the range of 300 to 700 nm. The K-T solvatochromic parameters—α (hydrogen bond donating ability), β (hydrogen bond accepting ability), and π* (dipolarity/polarizability)—using established equations based on the maximum absorption wavelengths (λ_{max}) of Nile red and 4-nitroaniline were calculated using the following equations [23].

$$\alpha = \frac{19.9675 - 1.0241 \times \pi - \nu_{(NR)max}}{1.6078} \quad (1)$$

$$\beta = 11.134 - \frac{3580}{\lambda_{(NH2)max}} - 1.125^{*}\pi \quad (2)$$

$$\pi = \frac{19.839 - \nu_{NR}}{2.9912} \quad (3)$$

$$\nu_{NR} = \frac{10,000}{\lambda_{(NR)max}} \quad (4)$$

2.4. Pretreatment process

In our biomass pretreatment method, a specific amount of tea stem powder (1.0 g) with 20.0 g of PDES were mixed in a 100 mL high-pressure reactor which was stirred and subjected to controlled

temperatures for predetermined times to complete the pretreatment. Afterward, the cooled mixture was separated via vacuum filtration, resulting in a cellulose-rich solid fraction. This solid fraction underwent further purification by being washed with 150 mL of an ethanol-water solution (1:1 ratio) and then dried at 80 °C until a constant weight was achieved. Meanwhile, the filtrate containing lignin-enriched PDES was reserved. Ethanol was removed from the filtrate through rotary evaporation, followed by dilution with 200 mL of deionized water to trigger lignin precipitation. The precipitated lignin was freeze-dried over 24 h and stored at 4 °C for future use.

To analyze the compositional changes resulting from the pretreatment, the cellulose, hemicellulose, and lignin contents in both the untreated and treated materials were determined following the protocols outlined by the National Renewable Energy Laboratory (NREL) [24]. The untreated tea stem consisted of 21 % cellulose, 23 % hemicellulose, and 37 % lignin. The effectiveness of the pretreatment was evaluated by measuring the retention of cellulose and the removal efficiencies of hemicellulose and lignin.

$$\text{Cellulose retention}\ (\%) = \frac{\text{Cellulose amount in solid residue}\ (g)}{\text{Initial amount of cellulose in the materials}\ (g)} \times 100\% \quad (5)$$

$$\text{Hemicellulose removal}\ (\%) = \left[1 - \frac{\text{Amount of hemicellulose in solid residue}\ (g)}{\text{Initial amount of hemicellulose in the materials}(g)}\right] \times 100\% \quad (6)$$

$$\text{Lignin removal}\ (\%) = \left[1 - \frac{\text{Amount of lignin in solid residue}\ (g)}{\text{Initial amount of lignin in the materials}(g)}\right] \times 100\% \quad (7)$$

MWL was isolated and purified by extracting ball-milled tea stem powder using a 1,4-dioxane solution, following a method based on established protocols [25].

2.5. Conversion process

Following a previous procedure [26], 6.0 g deionized water, 0.2 g of either tea stem or cellulose-rich residue with 1.5 g of the catalyst [C_4SO_3Hmim][$PhSO_3$] were mixed in a 35 mL high-pressure vial. The vial was then heated to 190 °C and held at this temperature for 4 h to promote the reaction. The resulting LA was determined using high-performance liquid chromatography (HPLC) analysis.

2.6. Preparation and application of fluorescent lignin

200 mg of regenerated lignin was dispersed in 40 mL 10 % H_2O_2 solution. This mixture was then heated to 120 °C in an oil bath for 6 h, resulting in a fluorescent lignin solution, which was subjected to freeze-drying for 48 h. The obtained fluorescent lignin was later used to detect metal ions. In this procedure, 0.3 mL of a metal ion solution with a concentration of 1600 μM was added to 3 mL of the fluorescent lignin solution, which had a concentration of 1 mg/mL. A fluorescence spectrophotometer (Fluoromax+, HORIBA Scientific, Japan) was used to examine the photoluminescence properties of the solution.

2.7. Characterization methods

HPLC was conducted using a Chromaster 5000 system from Hitachi Scientific Instruments Co. LTD. The system was equipped with a Bio-rad Aminex HPX-87H column, and a differential refractive index detector maintained at 50 °C for the analysis duration. A 5 mM sulfuric acid solution was used as the mobile phase, with a flow rate of 0.6 mL/min. X-ray diffraction (XRD) analysis was carried out using a Bruker D8 ADVANCE diffractometer, covering diffraction angles from 5° to 80° at a scanning speed of 8°/min. Fourier-transform infrared (FTIR) spectra were obtained with a Nicolet Is50 FTIR spectrometer, equipped with a mercury-cadmium-telluride (MCT) detector and an attenuated total reflectance (ATR) accessory. Gel permeation chromatography (GPC) was used to determine the molecular weight distribution of lignin, employing a Waters 1515 system coupled with an Agilent PLgel 5 μm MIXED-C column and a differential refractive index detector. Particle size distribution was assessed using a Malvern Zetasizer Nano ZS90, a laser-based particle size analyzer. Two-dimensional heteronuclear single quantum coherence (2D-HSQC) nuclear magnetic resonance (NMR) spectroscopy was performed on an AVANCE III 600 MHz spectrometer from Bruker, with the lignin sample dissolved in deuterated dimethyl sulfoxide.

2.8. Quantum chemical calculation

Lignin, a complex phenolic polymer composed of the three primary alcohol monomers: p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol, presents substantial challenges for structural simulation due to its intricate nature. Existing literature widely applied biomass monomers to do the DFT calculations [27,28]. In our study, p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol were chosen as representative lignin monomers for DFT analysis. To obtain the geometric structures of PDES-lignin interaction complexes, DFT calculations were conducted using the Gaussian 16 software package [29] based on the M062X/6-311 g(d,p) method. The energy minima of these optimized complexes were confirmed by ensuring that no imaginary frequencies were present, indicating stable configurations. Visualization of the reduced density gradient (RDG) for the interaction complexes was performed with VMD 1.9.4, which utilized data generated by Multiwfn 3.8 [30,31]. Simultaneously, a detailed analysis of atomic interactions within the molecules was carried out based on the atoms in molecules (AIM) theory [32]. Key properties, including electron density (ρ), the Laplacian of electron density ($\nabla^2\rho$), and energy density (H) at the bond critical point (BCP), were derived from the Multiwfn 3.8 output, providing valuable insights into the nature of the intermolecular interactions.

3. Results and discussion

3.1. Delignification efficiency by deep eutectic solvent

The pretreatment experiments were conducted using a mixture of HBA (PEG200) and HBD (organic acid) at a molar ratio of 2:1, under a temperature of 100 °C for 3 h. All the Brønsted acid-type HBD tested (levulinic acid, oxalic acid, formic acid, and lactic acid) were able to form homogenous PDES with PEG200. The pretreatment efficiency, as measured by cellulose retention, hemicellulose removal, and lignin removal across different PDES systems, was summarized in Fig. 1A. The

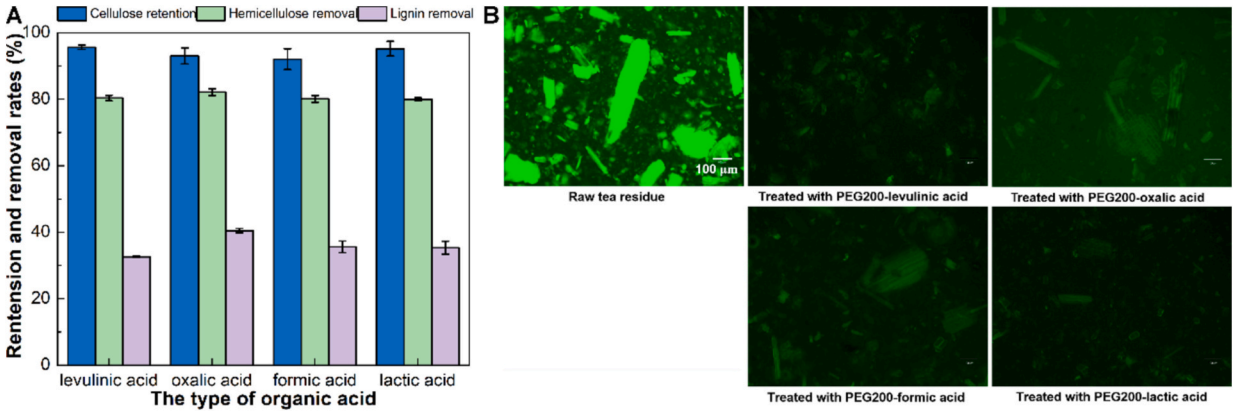


Fig. 1. The pretreatment efficiencies of different PEG200-organic acid PDES systems on tea stem (A) and the fluorescence microscopic images of the raw tea stem and raw tea stem pretreatment by different PEG200-organic acid PDES systems (B).

untreated tea stem consisted of 21 % cellulose, 23 % hemicellulose, and 37 % lignin. The results indicated that PEG200-based Brønsted acid PDESs were effective in delignifying tea stem, achieving successful removal of lignin and hemicellulose. In this process, lignin was extracted into the PDES, while the hemicellulose was removed primarily due to the conversion into valuable intermediates such as furfural and LA under acidic conditions. Among the PDESs, PEG200-OA exhibited the highest delignification efficiency (40.49 %), effectively dissolving lignin while retaining most of the cellulose. This efficiency can be attributed to the ample acid sites and appropriate acidity of the PEG200-OA PDES, which facilitated lignin dissolution and hemicellulose degradation. Besides, PEG200, as a hydrogen bond donor, has low viscosity and excellent solvent properties, allowing it to effectively dissolve various biomass components. Oxalic acid, serving as a hydrogen bond acceptor, helps to form a stable DES with PEG200, enhancing the solubility of lignocellulosic materials. This combination could disrupt the structural integrity of lignocellulose by weakening hydrogen bonds, facilitating the selective separation of lignin and hemicellulose. Consequently, PEG200-OA was selected as the optimal DES for subsequent experiments.

Lignin, an amorphous polymer composed of methoxylated phenylpropane units linked with phenolic groups [33], was known for emitting strong fluorescence across the visible and far-infrared spectra [34]. It was hypothesized that the fluorescence emitted by lignin in a solvent can serve as an indicator of its dissolution state, which correlated with the deconstruction of tea stem and the material's decrystallization [34]. To investigate this, fluorescence microscopy was employed to monitor lignin's dissolution dynamics in the different PDES systems, as depicted in Fig. 1B. Initially, intense fluorescence was observed in the tea stem solids, with no fluorescence detected in the solvent phase, indicating that lignin was densely encapsulating the tea stem solids. As the dissolution progressed in the various PDES systems, fluorescence began to appear in the solution, demonstrating effective lignin removal. The PEG200-OA system exhibited significantly higher fluorescence intensity in the solution compared to other systems, indicating greater lignin extraction. Thus, tracking fluorescence intensity within the mixture proved to be a reliable method for monitoring tea stem dissolution and highlighted the superior lignin extraction capability of the PEG200-OA treatment.

To further analyze the solvent's performance, Kamlet-Taft parameters were determined to assess the influence of solvent hydrogen bonding on lignin removal [23]. The parameters α, β, and π* corresponded to the solvent's hydrogen bond donating ability (hydrogen bond acidity), hydrogen bond accepting ability (hydrogen bond basicity), and polarizability, respectively [35]. The Kamlet-Taft parameters for the different PDES systems, along with the lignin removal rates, were presented in Table 1. By correlating the Kamlet-Taft parameters with lignin removal rates, it was observed that PEG200-OA PDES characterized by a lower β value and higher α, π*, and α-β values achieved higher lignin removal rates. These findings suggested that a higher net hydrogen donating ability (α-β) enhanced lignin dissolution. The α-β value indicated the PDES's ability to attack the β-O-4′ aryl ether bonds in lignin. PDESs with a high α-β value were more likely to form strong hydrogen bonds with lignin, breaking the β-O-4′ aryl ether bonds and promoting lignin degradation [36]. The Kamlet-Taft parameter results also confirmed that the PEG200-OA system exhibited superior lignin removal capability compared to the other PDES systems.

Table 1
The Kamlet−Taft parameters of the PDE.

PDES	α	β	π*	α-β	Lignin removal rate (%)
PEG200-oxalic acid	1.33	0.66	1.02	0.67	40.09
PEG200-formic acid	1.24	0.76	0.95	0.48	35.60
PEG200-lactic acid	1.24	0.77	0.95	0.47	35.35
PEG200-levulinic acid	1.23	0.74	0.94	0.49	32.62

3.2. Optimization of experimental conditions

To achieve optimal efficiency during the PEG200-OA PDES pretreatment, several key parameters, including reaction temperature, duration, the molar ratio of HBA to HBD, and water content were optimized. Among these, reaction temperature and duration emerged as critical factors influencing the pretreatment outcome. As depicted in Fig. 2A and B, the retention of cellulose within the substrate fluctuated with the reaction temperature, reaching its peak at 120 °C as the temperature increased from 100 °C to 150 °C. Hemicellulose removal exceeded 90 % when temperatures surpassed 130 °C, likely due to enhanced solvent-induced breakdown of tea stem components at elevated temperatures [37]. The lignin removal rate also rose significantly with increasing temperature, stabilizing beyond 140 °C. This stabilization could be attributed to the formation of pseudolignin, resulting from the recombination of small molecular fragments in an acidic environment as temperatures rise [38,39]. Considering the composition of the cellulose-rich substrate, the lignin removal rate, and energy consumption, a pretreatment temperature of 140 °C was deemed optimal. This temperature balanced low energy requirements with the highest observed lignin removal rate.

Fig. 2B further illustrated that a 4-h pretreatment duration resulted in a lignin removal rate of 58.30 %. However, extending the pretreatment time beyond this point led to a notable decline in lignin removal efficiency, possibly due to pseudolignin formation from tea stem [38,39]. Hemicellulose removal continued to increase up to 4 h, after which it plateaued. The ongoing degradation of hemicellulose during this period can impact the recovery and reuse of the solvent system. Thus, selecting a 4-h pretreatment period ensured efficient lignin

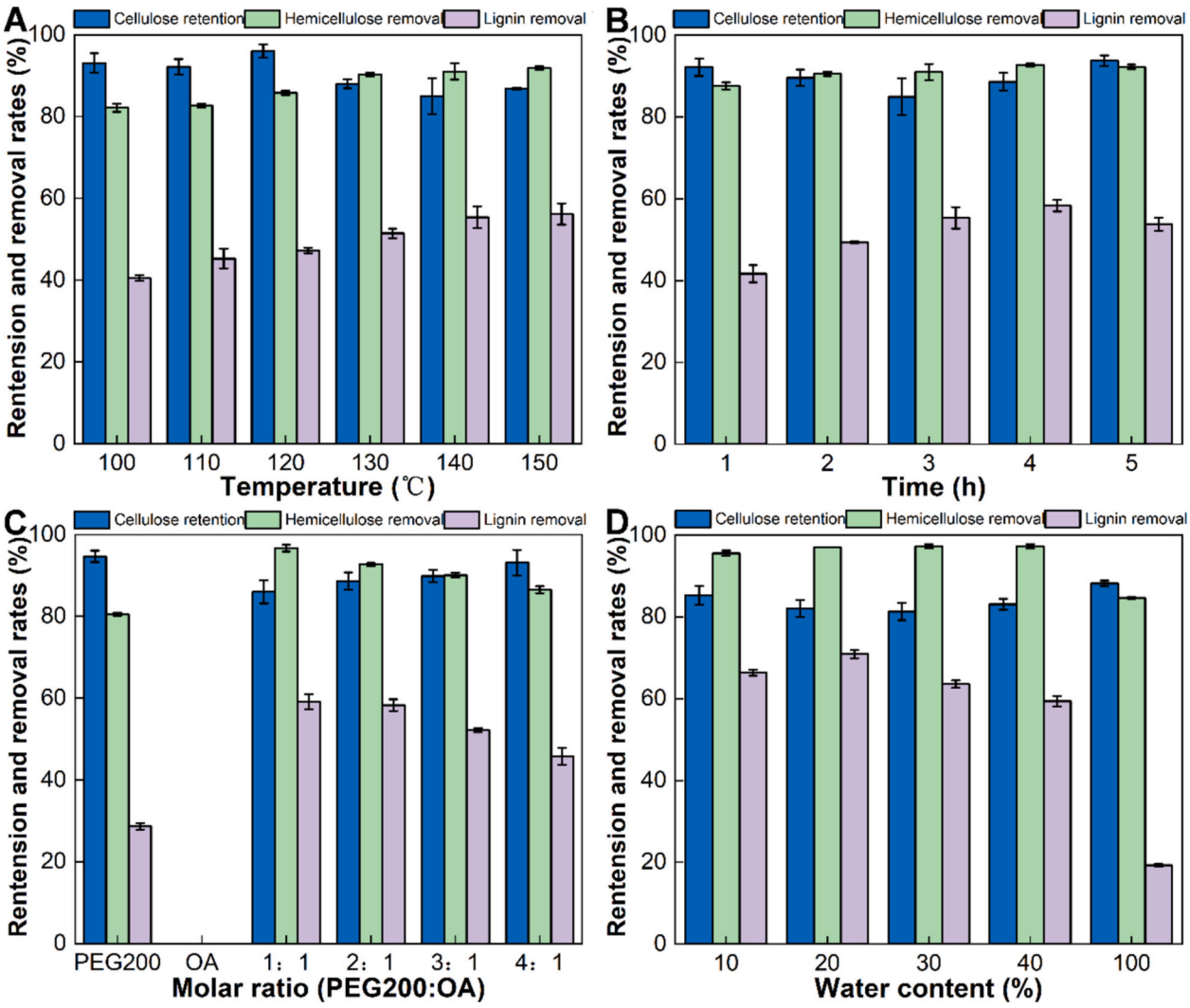


Fig. 2. The effects of reaction temperature (A: 1 g substrate, 20 g PEG200-OA (2:1), 3 h reaction time), reaction time (B: 1 g substrate, 20 g PEG200-OA (2:1), 140 °C reaction temperature), the molar ratio of PEG100 to OA (C: 1 g substrate, 20 g PEG200-OA PDES with 140 °C reaction temperature and 4 h reaction time) and the amount of water (D: 1 g substrate, 20 g solvent, PEG200-OA (2:1) with 140 °C reaction temperature and 4 h reaction time) on the pretreatment efficiency.

removal, minimized tea cellulose loss, and reduced pretreatment costs.

The influence of PDES components, specifically the varying molar ratios of PEG200 to OA, on lignin fractionation efficiency was systematically explored. Fig. 2C highlighted that pure OA struggled to mix effectively with the substrate, rendering it ineffective for pretreating tea stem. Conversely, pure PEG200 exhibited a high cellulose retention rate but demonstrated limited efficiency in lignin removal. As OA was introduced into the mixture, the cellulose retention rate decreased while the hemicellulose removal rate increased, a result of the conversion processes occurring in the acidic environment. Adjusting the PEG200 to OA ratios from 1:1 to 4:1 led to a reduction in delignification efficiency, dropping from 59.11 % to 45.80 %. This result indicated a synergistic interaction between PEG200 and the Brønsted acid present. The Brønsted acid, consisting of small ions, enhanced the swelling of tea stem biomass, thereby improving the PDES's access to lignin-carbohydrate complexes (LCCs). Additionally, the PEG200 stabilized the PDES system by modulating its acidity. However, when the PEG200 content was reduced to a 1:1 ratio with OA, the system's stability was compromised, and PDES formation was ineffective. Thus, a PEG200 to OA molar ratio of 2:1 was identified as the optimal balance, ensuring efficient lignin removal, minimal tea cellulose loss, and a stable PDES solvent system.

Beyond the molar ratio, water content also played a critical role in the efficiency of the pretreatment process. Introducing water reduced the system's viscosity, enhancing the interaction between the substrate and solvent and improving mixture uniformity, which also helped to lower costs. However, it was crucial to consider that water addition may dilute the PDES concentration. As shown in Fig. 2D, the pretreatment efficiency was significantly affected by water content. Hemicellulose and lignin removal rates increased when water content reached 10 %, while further increases in water content did not significantly alter cellulose retention or hemicellulose removal. The lignin removal rate peaked at 20 % water content. Consequently, for both cost-effectiveness and process simplicity, a water content of 20 % was chosen.

In summary, the optimal conditions for the pretreatment process were identified as a temperature of 140 °C, a duration of 4 h, a water content of 20 %, and a PEG200 to OA molar ratio of 2:1. Under these conditions, 82.03 % of cellulose was retained in the solid residues, with 97.02 % of hemicellulose and 70.89 % of lignin successfully removed from the tea stem.

3.3. Characterization and utilization of the solid residues

3.3.1. Characterization of the solid residues

To gain deeper insights into the underlying pretreatment mechanism, FTIR spectroscopy and XRD analyses were conducted on untreated tea stem and cellulose -rich solid residues obtained at optimized

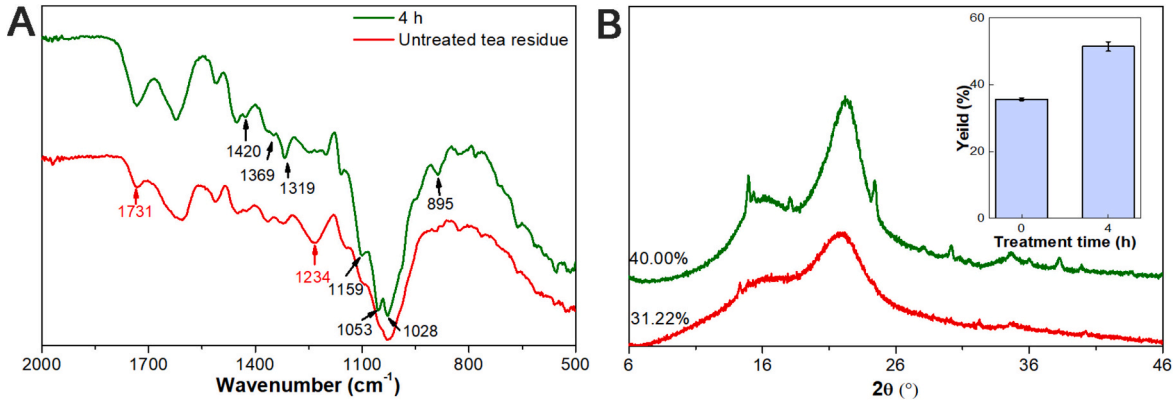


Fig. 3. The FTIR (A) and XRD (B) spectra of untreated tea stem and cellulose-rich residue under the optimized reaction condition. The insert figure was the conversion yield of LA.

pretreatment condition. The FTIR analysis, shown in Fig. 3A, provided a detailed examination of the structural changes occurring during pretreatment with PEG200-OA. The most significant spectral changes in cellulose, hemicellulose, and lignin components were observed below 2000 cm^{-1}. The peak at 1234 cm^{-1}, corresponding to the C – O stretching vibration within ester linkages between lignin and hemicellulose [40–42], disappeared following treatment. This disappearance indicated the removal of hemicellulose and lignin from the tea stem during PEG200-OA pretreatment. After pretreatment, there was a marked increase in the intensity of peaks at 1420 cm^{-1}, 1369 cm^{-1}, 1319 cm^{-1}, 1159 cm^{-1}, 1053 cm^{-1}, and 1028 cm^{-1}, which corresponded to various vibrations such as C – O – C stretching, C – H2 deformation, C – H2 rocking, C – O – C glycoside ether, C – O pyranose ring stretching, and C – O stretching of a primary alcohol [40–42], all of which were characteristic of cellulose in carbohydrates. The distinct band at around 895 cm^{-1} was attributed to the stretching vibrations of C – O – C, signifying the presence of β-glycosidic bonds within cellulose [40–42]. The enhanced of these bands after treatment indicated the enrichment of cellulose in the solid residues. In the untreated tea stem, a spectral band around 1731 cm^{-1}, indicative of carbonyl groups (C=O), was observed in both lignin and hemicellulose [40–42]. The persistence of this band in the cellulose-rich solid residue suggested that carbonyl stretching vibrations, primarily due to the esterification of OA with cellulose hydroxyl groups, remained present post-treatment.

Fig. 3B showed the XRD patterns of both untreated tea stem and cellulose-rich residue. After treatment, the solid residues showed increased peak intensity at 16.5° and 34.5°, indicative of a strengthened cellulose I structure for the (004) and (110) lattice planes [43,44]. The cellulose crystalline structure remained intact, retaining its cellulose I form post-treatment, suggesting that PEG200-OA primarily acted on lignin and hemicellulose without affecting cellulose crystallinity. The crystallinity index (CrI) was calculated for each sample, revealing a notable increase in the crystallinity of the solid residues after treatment. This increase was likely due to the removal of amorphous components, such as lignin and hemicellulose, during the pretreatment process [45]. These results align with previous studies that had demonstrated an increase in crystallinity following lignin and hemicellulose removal through pretreatment.

3.3.2. *Utilization of the cellulose-rich solid residues*

Extensive research had been dedicated to transforming biomass into valuable intermediates, with particular emphasis on LA due to its versatility and potential applications. As lignin is a highly recalcitrant polymer, its presence in the biomass matrix significantly hinders the accessibility of cellulose to catalytic action, thus reducing the efficiency of cellulose conversion. Lignin removal helps to disrupt the biomass structure, improving the porosity and accessibility of cellulose, which in turn enhances its conversion to levulinic acid. In this study, we effectively removed lignin during the pretreatment process. As illustrated in Fig. 3B, LA production markedly increased from 35.7 % to 51.5 % following PEG200-OA pretreatment. Thus, pretreating tea stem with PEG200-OA significantly enhanced the conversion of cellulose into LA, optimizing the overall process.

Therefore, the PEG200-OA pretreatment successfully removed lignin and hemicellulose without compromising the integrity of the cellulose structure. The elimination of amorphous lignin led to an increase in the crystallinity of the residue, thereby improving the efficiency of cellulose conversion into LA.

3.4. *Characterization of the regenerated lignin*

3.4.1. *Particle size, molecular weight, and chemical structure of regenerated lignin*

The characteristics of regenerated lignin, including particle size, molecular weight, and chemical structure, were analyzed using a laser particle size analyzer, GPC, and 2D HSQC NMR spectroscopy. These findings were compared with those of MWL to gain deeper insights into the mechanisms of lignin extraction.

As shown in Fig. 4A, the PEG200-OA treatment resulted in the production of nanometer-scale lignin particles with smaller and more uniformly distributed sizes compared to MWL. This result indicated that PEG200-OA treatment effectively broke down lignin macromolecules into simpler structure that were more amenable to further processing and modification. The nanometer-scale lignin also showed promise for various applications [46]. During the PDES treatment, the chemical structure of lignin was altered, impacting its molecular weight. The weight-average (Mw) and number-average (Mn) molecular weights were shown in Fig. 4B. The extracted lignin from tea stem after PEG200-OA DES treatment exhibited Mw and Mn value of 1283 g/mol and 531 g/mol, respectively. Notably, these molecular weights were significantly lower than those of MWL.

The structural characteristics of lignin regenerated from PEG200-OA PDES pretreatment were examined using 2D-HSQC spectroscopy (Fig. 5). The 2D-HSQC spectrum provided insights into the lignin macromolecule substructures through the side-chain region ($\delta C/\delta_H$ 50–95/2.5–6.0) and detailed the lignin's structural units via the aromatic region ($\delta C/\delta_H$ 95–150/5.6–8.0) [33,47,48]. The signal assignments in the 2D-HSQC spectra were made based on previous studies [33,47,48] and were listed in Table S1 in Supporting Information.

The lignin side-chain region revealed the presence of β-aryl ether linkages (β-O-4, A), resinol structures (β-β′, B), and phenylcoumaran structures (β-5, C), with β-O-4 being the dominant linkage, significantly influencing lignin's physicochemical properties [33,47,48]. Comparative analysis of MWL (Fig. 5A) showed distinct signals corresponding to

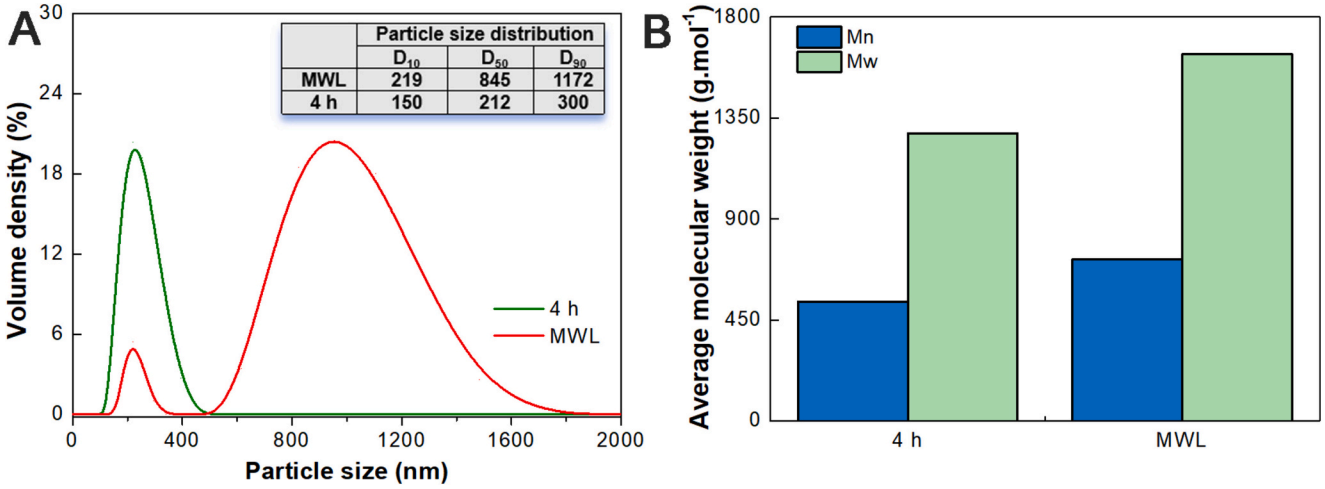


Fig. 4. The particle size distributions (A) and molecular weights (B) of the MWL and the regenerated lignin isolated under the optimized reaction condition.

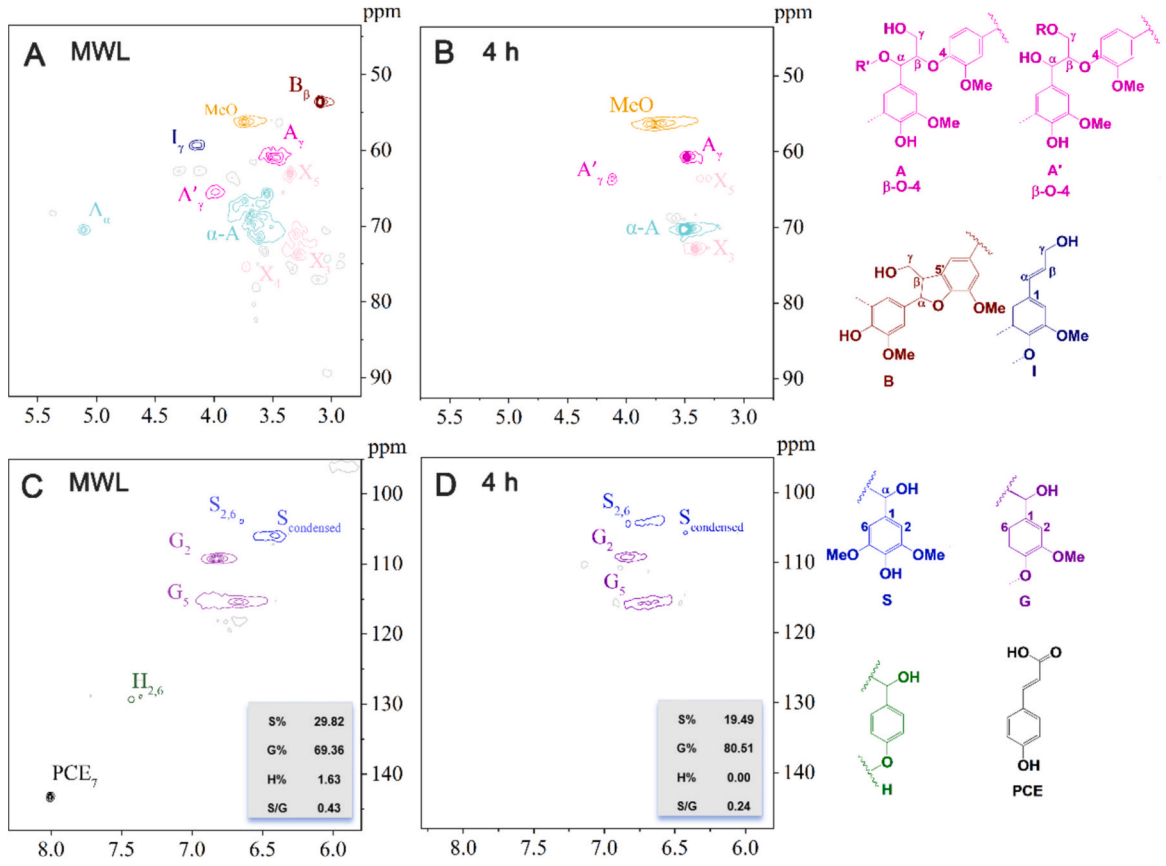


Fig. 5. The side-chain region (A and B) and the aromatic region of 2D HSQC NMR spectra of MWL and regenerated lignin isolated under the optimized reaction condition.

β-O-4 and β-5 linkages. The β-O-4 content in MWL, measured at 40.00/100Ar, was notably reduced following PEG200-OA pretreatment, while β-β′ signals were absent post-treatment. These observations suggest that depolymerization of lignin occurred during the pretreatment process. Additionally, both MWL and regenerated lignin exhibited signals associated with xylan (X), though signals corresponding to X4 and X5 disappeared after pretreatment, indicating effective xylan removal. Thus, the pretreatment effectively preserved cellulose, cleaved β-O-4 linkages, and disrupted lignin-polysaccharide bonds, leading to lignin removal. Moreover, pronounced methoxy (–OCH_3) signals were detected, likely due to esterification between lignin hydroxyl groups and OA.

In the aromatic region (Fig. 5C and D), signals from hydroxyphenyl (H), guaiacyl (G), and syringol (S) units, along with p-coumarates (PCE) structures in tea lignin, were observed. Notably, signals related to G and S units were particularly intense. MWL showed S and G unit contents of 29.82 % and 69.36 %, respectively. After pretreatment, H and PCE signals were substantially disappeared, with S unit content decreasing to 19.49 % and G unit content increasing to 80.51 %. This suggested that G units were more prone to condensation than S units during pretreatment. The S/G ratio decreased from 0.43 in MWL to 0.24, indicating a

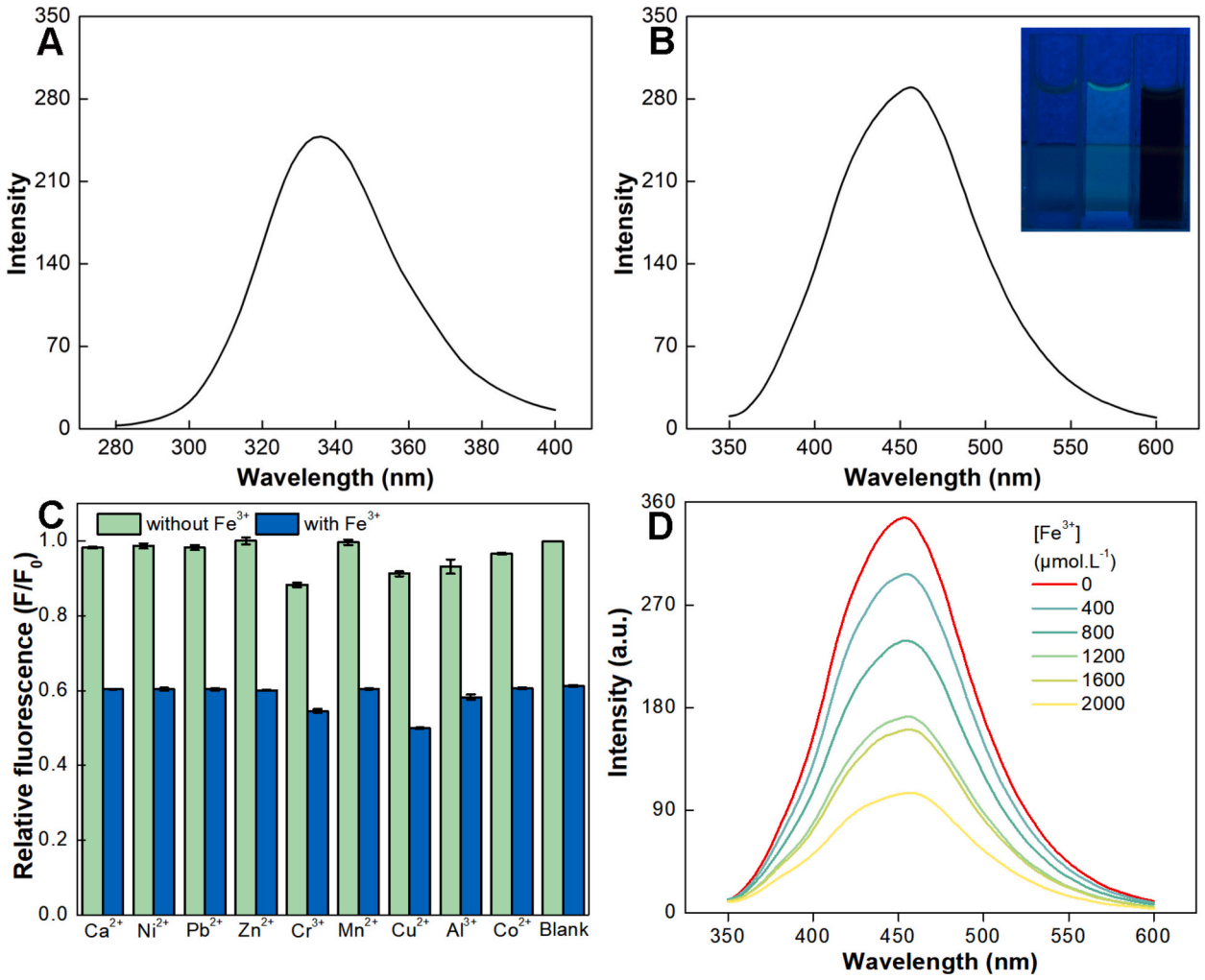


Fig. 6. A: the fluorescence excitation spectrum of fluorescent lignin; B: the emission spectrum of fluorescent lignin. The inset images from left to right were water, fluorescent lignin and regenerated lignin dissolved in NaOH solution under UV light at 365 nm; C: the relative loss of photoluminescence intensities of fluorescent lignin aqueous solutions with the presence of individual interference metal ions, or with the presence of both individual metal ions and Fe^{3+} ions; D: photoluminescence spectra of fluorescent lignin at various Fe^{3+} concentrations.

selective degradation or removal of S-type lignin units during the pretreatment process.

3.4.2. *Photoluminescence properties and applications of fluorescent lignin*

The fluorescent lignin was simply prepared via oxidized regenerated lignin by the H_2O_2 solution. The fluorescent lignin solution was found to have an excitation wavelength around 340 nm (Fig. 6A), with an emission wavelength near 450 nm when excited at 340 nm (Fig. 6B). Under UV light at 365 nm (inset of Fig. 6B), the lignin exhibited strong photoluminescence, emitting a bright fluorescence that indicated good dispersion and uniformity, making it suitable for various applications.

Comparison between the fluorescent lignin prepared from the regenerated lignin and that prepared from MWL was also conducted. The results were present in Fig. S1 in Supporting Information. The fluorescent lignin derived from regenerated lignin exhibited significantly higher fluorescence intensity than the fluorescent lignin obtained from MWL. This observation highlighted the distinct advantages of the regenerated lignin from PDES in terms of its photoluminescence properties, which could offer improved performance for applications in fluorescence-based sensing or detection.

Heavy metal ions pose significant environmental and health risks, making the development of sensing technologies for detecting these ions crucial. Due to the high quenching effect of heavy metal ions, fluorescent lignin had demonstrated excellent detection capabilities for various metal ions [49,50]. The influence of the pH on the fluorescence intensity was tested before metal ions detection. In Fig. S2 (Supporting information), as the pH increased, the fluorescence intensity initially increased and then decreased, with the maximum fluorescence intensity observed at pH = 6. This finding highlighted the significant influence of pH on the Fe^{3+} detection process, which may be attributed to changes in the interaction between Fe^{3+} ions and the fluorescence probe as well as the protonation state of the probe itself. On the other hand, when the pH value changed from 4 to 7, the fluorescence intensity of the fluorescent lignin was only minimally affected. The pH values of various metal ions also felt within this range, so their impact on subsequent metal ion detection was negligible. As shown in Fig. 6C, the fluorescent lignin solution was tested against different metal ions. The addition of Fe^{3+} resulted in the most pronounced fluorescence quenching, while other metal ions did not cause significant changes in fluorescence intensity, suggesting that the prepared fluorescent lignin was particularly effective as a probe for detecting iron ions.

To evaluate the selectivity of the fluorescent lignin for Fe^{3+}, we tested it against Ca^{2+}, Ni^{2+}, Pb^{2+}, Zn^{2+}, Cr^{3+}, Mn^{2+}, Cu^{2+}, Al^{3+}, and Co^{2+} as competing metal ions. The results in Fig. 6C showed that the quenching response to Fe^{3+} was not significantly affected by the presence of these other ions. The high selectivity of the fluorescent lignin sensor for Fe^{3+} was likely due to Fe^{3+} having a higher thermodynamic affinity and a faster chelation process with fluorescent lignin compared

to other metal ions. This interaction likely facilitated the formation of Fe-lignin complexes, promoting charge transfer and inhibiting exciton recombination, leading to the observed photoluminescence quenching.

Further investigation into the sensitivity and feasibility of using fluorescent lignin for quantitative Fe^{3+} detection revealed that, as shown in Fig. 6D, increasing concentrations of Fe^{3+} resulted in a gradual decrease in fluorescence intensity, confirming the effective quenching of lignin fluorescence by Fe^{3+}. The fluorescent lignin demonstrated a clear dependency on Fe^{3+} concentration.

These findings suggested that the novel fluorescent sensor developed in this study held potential for the determination of Fe^{3+} in aqueous environments. The fluorescence-based methods for detecting metal ions with fluorescent lignin offered advantages such as low cost, high sensitivity and selectivity, and rapid response times.

3.5. Calculation and analysis of intermolecular interaction by DFT

In the previous section, the PEG200-OA PDES system demonstrated considerable effectiveness in lignocellulose pretreatment, particularly in its ability to dissolve lignin. This effectiveness was closely related to the interaction between the PDES components and lignin. To better understand the mechanisms underlying lignin dissolution, we conducted DFT calculations to analyze the interactions between the PEG200-OA system and lignin [51,52]. The geometries of the interaction complexes between PEG200-OA and p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol were optimized, with the most stable structures shown in Fig. 7.

Fig. 7 illustrated the weak intermolecular interactions identified using the RDG function isosurface (top) and scatter diagram (bottom). Based on the Sign $(\lambda^2)\rho$ values, these interactions were categorized into hydrogen bonds (blue), van der Waals forces (green), and steric effects (red) [31]. Our analysis revealed that hydrogen bonds and van der Waals forces were the primary interactions between PEG200-OA and lignin, with both the HBA and HBD playing a synergistic role in lignin removal. Notably, the hydrogen bonds in the PEG200-OA-lignin complex were formed between the hydrogen atom on the carboxyl group of the HBD (OA) and the oxygen atom on the alkyl hydroxyl group of lignin.

The properties of these hydrogen bonds were further elucidated through topological analysis using AIM theory, as summarized in Fig. 7. It was found that the ρ at the BCP of a hydrogen bond typically ranged from 0.002 to 0.035 au, while the $\nabla^2\rho$ at the BCP of non-covalent interactions, such as hydrogen bonds, generally fell within the range of 0.024 to 0.139 au. The values of ρ and $\nabla^2\rho$ for the hydrogen bonds between PEG200 and OA fell within these ranges, confirming the formation of hydrogen bonds.

According to established theories [32,53], the strength and nature of hydrogen bonds can be determined by specific criteria. If both the $\nabla^2\rho$ and the *H* were negative, the hydrogen bond is considered strong, shared, and covalently dominant. Conversely, if $\nabla^2\rho$ was positive and *H* was negative, the bond was of medium strength, closed-shell, and covalently dominant. In cases where both $\nabla^2\rho$ and *H* are positive, the hydrogen bond was classified as weak, closed-shell, and electrostatic dominant. In our study, the positive values of $\nabla^2\rho$ and *H* indicated that the interactions between PEG200-OA and lignin were weak, closed-shell, and electrostatic in nature.

4. Conclusion

In this study, PDESs were developed to separate cellulose, hemicellulose and lignin from tea stem, facilitating the production of platform molecule and fluorescent lignin. The combination of PEG200 and OA in the PDESs resulted in a high cellulose retention rate (82.03 %) and the most effective lignin removal (70.89 %). Both PEG200 and OA significantly contributed to lignin dissolution, with weak interactions between DES and lignin driven mainly by hydrogen bonding (characterized as

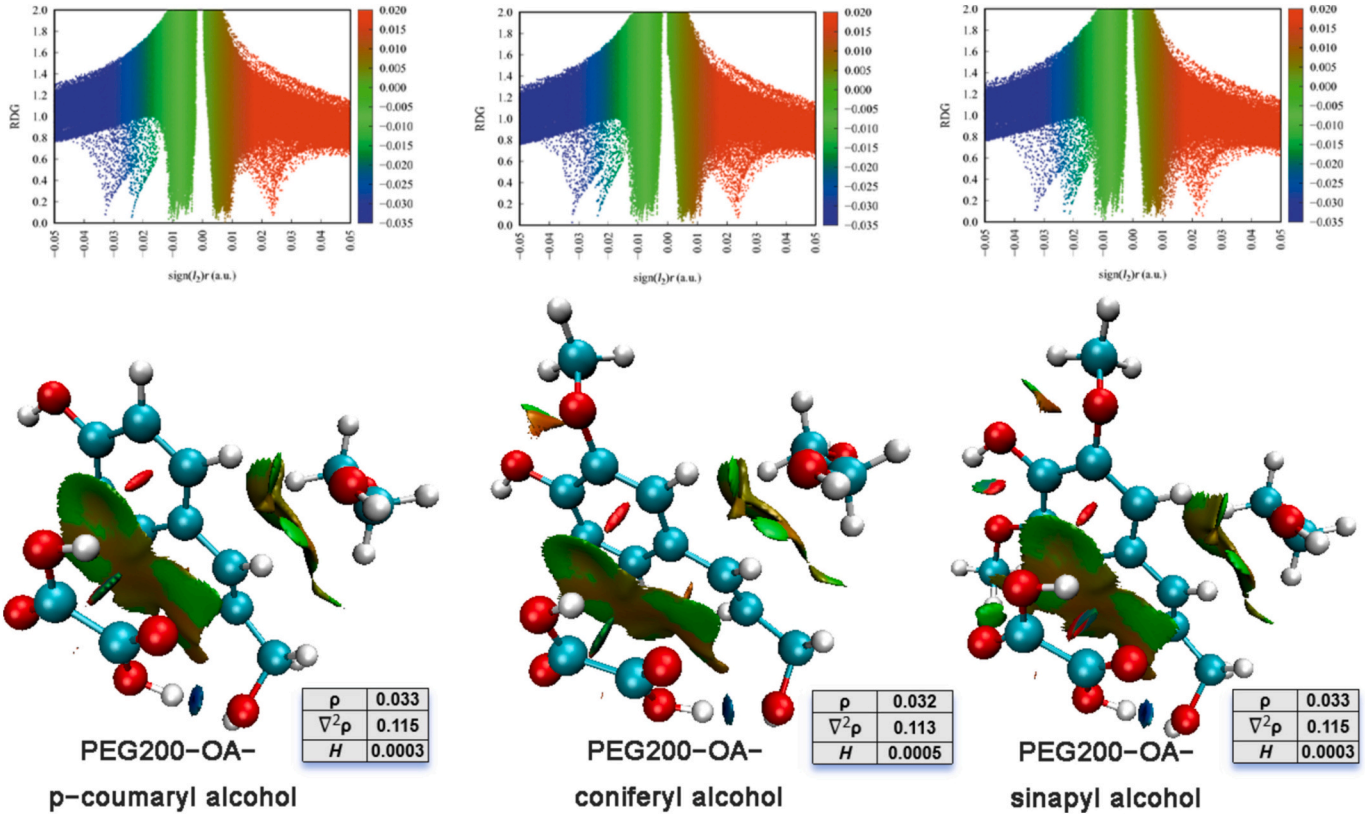


Fig. 7. RDG analysis of the PEG200-OA-lignin interaction complex.

weak, closed-shell, and electrostatic) and van der Waals forces. The pretreatment with PEG200-OA led to a substantial increase in platform molecule LA production, from 35.7 % to 51.5 %. The regenerated lignin exhibited lower molecular weight and nanoscale particle size compared to MWL, indicating its potential for use in lignin-based chemicals and materials. This reduction in molecular weight also facilitated the oxidation process, enabling the preparation of fluorescent lignin for metal ion detection. Overall, this study introduced a novel, environmentally friendly solvent system for biomass pretreatment in biorefinery applications, enhancing the value of agricultural waste.

CRediT authorship contribution statement

Pei-Ru Zhou: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiao-Ping Zheng:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Ya-Peng Du:** Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yu Chai:** Writing – review & editing, Visualization, Software, Resources, Methodology, Investigation, Formal analysis. **Yu-Cang Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Yan-Zhen Zheng:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.139922.

Data availability

Data will be made available on request.

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